

```
1 #include <stdio.h>
2
3 #define MAX 20
4 #define INF 1000000000
5
6 int main() {
7     int n, g[MAX][MAX], i, j, s;
8     if (scanf("%d", &n) != 1) return 0;
9
10    for (i = 0; i < n; i++)
11        for (j = 0; j < n; j++)
12            scanf("%d", &g[i][j]);
13
14    scanf("%d", &s);
15    s--;
16
17    int dist[MAX], vis[MAX];
18    for (i = 0; i < n; i++) {
19        dist[i] = g[s][i];
20        if (dist[i] >= INF) dist[i] = INF;
21        vis[i] = 0;
```

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Dijkstra's Algorithm

DIJKSTRA'S ALGORITHM

Develop a C program that accepts a graph as input and displays the shortest distance from a given source node to every other node using Dijkstra's algorithm.

PROGRAM EDITOR

C

Run Save

```
25 for (i = 0; i < n; i++) {  
26     int u = -1;  
27     for (j = 0; j < n; j++)  
28         if (!vis[j] && (u == -1 || dist[j] < dist[u]))  
29             u = j;  
30  
31     if (u == -1) break;  
32     vis[u] = 1;  
33  
34     for (j = 0; j < n; j++) {  
35         if (!vis[j] && g[u][j] < INF && dist[u] + g[u][j] < dist[j])  
36             dist[j] = dist[u] + g[u][j];  
37     }  
38 }  
39  
40 for (i = 0; i < n; i++) {  
41     if (dist[i] >= INF) printf("Node %d: INF\n", i + 1);  
42     else printf("Node %d: %d\n", i + 1, dist[i]);  
43 }  
44 return 0;  
45 }
```

INPUT

Your INPUT goes
here!